

DATP-CP

Research Dossier

Calibration-Channel Poisoning in Federated IoT Anomaly Detection

Version: v1.0

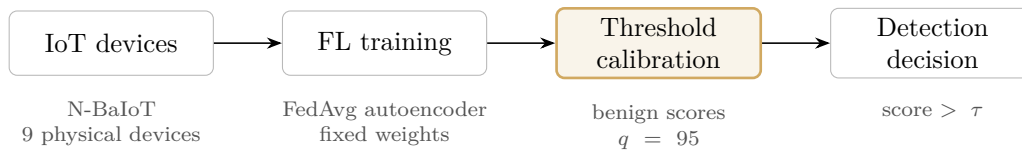
Status: Scientific and technical scoping document

Scope: N-BaIoT threshold-scope study. Attack surface: benign calibration scores only. Training data, model weights, aggregation, test scores, and labels are not modified.

Purpose

- Study the post-training threshold-calibration stage as an attack surface.
- Modify only benign calibration scores.
- Preserve training data, aggregation, model weights, test scores, and test labels.
- Compare GLOBAL_THRESHOLD, LOCAL_THRESHOLD, and CLUSTER_THRESHOLD.
- Keep claims bounded to the tested N-BaIoT protocol.

Minimal Pipeline



Only the *threshold calibration* step is within the attack surface.

Scope Boundary

- Calibration scores may change.
- Training data does not change.
- Model weights do not change.
- Aggregation does not change.
- Test scores and labels do not change.
- AUROC invariance is a sanity check only.

Project Identity

Field	Value
Project acronym	DATP-CP
Full title	Calibration-Channel Poisoning in Federated IoT Anomaly Detection
Document type	Research Dossier
Version	v1.0
Status	Scientific and technical scoping document
Thesis line	Collaborative Malware Detection with Federated Learning in IoT Networks
Research branch	Security / calibration-channel poisoning
Main dataset	N-BaIoT
Client definition	Physical IoT device as FL client
Model setting	Fixed FedAvg autoencoder
Threshold policies	GLOBAL_THRESHOLD, LOCAL_THRESHOLD, CLUSTER_THRESHOLD
Main claim scope	Single-client calibration compromise

Document Purpose

This document:

- Defines research scope.
- Defines scientific objectives.
- Defines methodological boundaries.
- Defines validation criteria.
- Defines expected deliverables.
- Defines risks and limitations.

This document is not:

- A paper draft.
- A presentation deck.
- An implementation log.

Context and Positioning

DATP Baseline

- DATP studied threshold calibration scope in non-IID federated IoT anomaly detection.
- It fixed the FedAvg autoencoder and varied only the threshold policy.
- It showed that threshold scope governs per-client false-alarm disparity.
- It established calibration as a decisive operating-point variable.

DATP-CP Rationale

- DATP-CP asks whether the same calibration stage is vulnerable.
- The attack occurs after training.
- Existing FL defenses usually focus on training, updates, or aggregation.
- DATP-CP studies the post-training calibration gap under strict causal isolation.

Three-Branch Comparison

	DATP	DATP-CP	Journal extension
Scientific role	Performance study	Security study	Generalization study
Main question	Which policy minimizes FPR disparity?	Does calibration poisoning produce policy-dependent harm?	Do findings generalize?
Primary variable	Threshold policy	Poison fraction f	Dataset / FL protocol
Adversary	None	Single compromised client	None
Claim boundary	N-BaIoT, threshold-scope study	N-BaIoT, single-client	External datasets

Key Distinctions

- **DATP:** performance personalization.
- **DATP-CP:** calibration-channel security.
- **Journal extension:** external validation and generalization.

Problem

- Thresholds are computed after training.
- Thresholds define the final operating point.
- Benign calibration scores determine τ .
- This stage is separate from model training and aggregation.
- The calibration buffer may become an overlooked attack surface.

Central Research Question

Does poisoning only benign threshold-calibration scores produce material, policy-dependent downstream harm under `GLOBAL_THRESHOLD`, `LOCAL_THRESHOLD`, and `CLUSTER_THRESHOLD`?

Objectives

Scientific objectives

- Isolate threshold calibration as an attack surface.
- Measure material threshold movement.
- Compare vulnerability across threshold policies.
- Separate raising and lowering failure modes.
- Preserve causal isolation.

Technical objectives

- Validate clean DATP-compatible artifacts.
- Implement controlled benign-score replacement.
- Enforce valid source-objective pairs.
- Run paired clean-vs-poisoned comparisons.
- Produce reproducible outputs.

Scope

Included	Excluded
N-BaIoT main study	Training-data poisoning
Physical device clients	Model poisoning
Fixed FedAvg autoencoder	Aggregation poisoning
Benign calibration-score poisoning	Backdoors
Score-level proxy attack	Evasion
<code>GLOBAL_THRESHOLD</code>	Privacy mechanisms
<code>LOCAL_THRESHOLD</code>	Deployment benchmarking
<code>CLUSTER_THRESHOLD</code>	Raw-traffic attack generation
$q = 95$	Journal-extension datasets as main evidence
Single-client main study	Model-personalization comparators
Paired clean-vs-poisoned evaluation	Universal FL robustness claims
<code>RANDOM_BENIGN</code> control	Privacy guarantee claims

Methodology and Validation

Experimental Setup

Dataset — N-BaIoT; 9 physical IoT devices; one device equals one FL client; $n_{\text{cal}} \geq 100$.

Model — Federated autoencoder; FedAvg; fixed model weights; clean and poisoned runs share the same model.

Calibration — Benign reconstruction scores; p95 threshold rule; $q = 95$; policies: GLOBAL, LOCAL, CLUSTER.

Attack Mechanism

1. Select a victim client.
2. Choose poison fraction f .
3. Select replacement positions in the benign calibration array.
4. Sample replacement values from the victim-local benign reservoir.
5. Recompute thresholds.
6. Evaluate on unchanged test scores and labels.

Source Strategies

- HIGH_SCORE_BENIGN $\rightarrow$ THRESHOLD_RAISE
- LOW_SCORE_BENIGN $\rightarrow$ THRESHOLD_LOWER
- RANDOM_BENIGN $\rightarrow$ negative control

Poison Fractions

$$f \in \{0.10, 0.20, 0.40\}$$

Validation Criteria

Criterion	Metric	Interpretation
Signed $\Delta\tau$	Direction	Attack in correct direction
Material $ \Delta\tau $	Magnitude	Effect exceeds materiality bound
ΔTPR	Detection drop	Under THRESHOLD_RAISE
$\Delta\text{CV}(\text{FPR})$	Alarm dispersion	Under THRESHOLD_LOWER
Policy differentiation	Differential response	Policies respond differently
RANDOM_BENIGN	Control audit	Negative-control baseline
AUROC invariance	Sanity check	Not a primary metric

Claim Gates

1. **Gate 1:** Material correctly signed $\Delta\tau$.
2. **Gate 2:** Directional effect exceeds RANDOM_BENIGN.
3. **Gate 3:** Downstream harm metric moves as expected.

Claim Classes

- **Full vulnerability:** All gates pass.
- **Mechanism-only:** Threshold movement only.
- **Calibration instability:** Negative control also moves.
- **Null or conditional:** Evidence insufficient.

Deliverables

- D1** Research protocol and scope definition.
- D2** Attack implementation and calibration-injection pipeline.
- D3** Clean artifact validation report.
- D4** Poisoning experiment outputs.
- D5** Figures and result tables.
- D6** Conference manuscript.
- D7** Reproducibility package.
- D8** Presentation dossier.

Risks and Limitations

- Score-level proxy limitation.
- N-BaIoT-only main evidence.
- Single-client compromise main claim.
- No deployment benchmark.
- AUROC invariance is a sanity check only.
- With-replacement sampling may duplicate scores.

Work Packages

- WP1** Scope lock and scientific boundary audit.
- WP2** Clean artifact validation.
- WP3** Attack implementation and validation tests.
- WP4** N-BaIoT main experiment execution.
- WP5** Analysis, figures, and tables.
- WP6** Manuscript preparation.
- WP7** Final audit and packaging.

Governance

- Keep claims bounded to the tested protocol.
- Disclose score-level proxy limitation.
- Treat AUROC invariance only as a sanity check.
- Do not convert optional extensions into main claims.
- Do not claim raw-traffic exploitability.
- Do not claim deployment readiness.

Research Contribution

DATP-CP contributes:

- A bounded calibration-channel threat model.
- Causal isolation from training and aggregation.
- Policy-differentiated vulnerability evidence.
- A bridge between DATP and adversarial robustness.